Reg. No. | | | | | | | | | | | [14CE33]

THIRD SEMESTER B.TECH. EXAMINATION, DECEMBER –2015

FLUID MECHANICS

Time: 3 Hours **Maximum Marks:70**

Answer any FIVE full questions choosing at least two from each part.

PART - A

1. a. Define the following terms with units: **06**
   - i. Mass density
   - ii. Weight density
   - iii. Relative density

   b. Calculate the capillary effect in millimetres in a glass tube of 2.5mm diameter, when immersed vertically in **08**
   - i. Water and
   - ii. Mercury.

   The value of the surface tension of water and mercury in contact with air are 0.0725N/m and 0.52 N/m respectively. The angle of contact for water is zero and that for mercury is 130°. The specific gravity for mercury is given 13.6.

2. a. State and derive "Hydro- Static law". Show the pressure head. **07**

   b. A Circular opening, 3m diameter, in a vertical side of a tank is closed by a disc of 3m diameter which can rotate about a horizontal diameter. Calculate: (i) the force on the disc in equilibrium in the vertical position when the head of water above the horizontal diameter is 6m. **07**

3. a. With a neat sketch, differentiate between simple manometer and differential manometer. **04**

   b. A single column vertical manometer (i.e, micrometer) is connected to pipe containing oil of specific gravity 0.9. The area of the reservoir is 80 times the area of the manometer tube. The reservoir contains mercury of specific gravity 13.6. the level of the mercury in the reservoir is at a height of 30cm below the center of the pipe and difference of mercury levels in the reservoir and right limb is 50cm. Find the pressure in the pipe. **10**

4. a. Derive an expression for continuity equation for a three-dimensional flow. **07**

   b. In a two-dimensional incompressible flow, the Fluid Velocity components are given by u=x-4y and v=-y-4x. Show that velocity potential exists and determine its form. **07**

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PART - B

5. a. Derive Euler's equation of motion. Obtain Bernoulli's equation from Euler's equation. List the assumptions made in it. **09**

   b. What are the major and minor losses in a pipe flow? **05**

6. a. Explain the term, 'dimensionally homogeneous equation. **03**

   b. A partially sub-merged body is towed in water. The resisting force F to its motion depends on the density ρ, the viscosity μ of water, length L of the body, velocity v of the body and the acceleration due to gravity g. Show that the resisting force to the motion can be expressed in the form, $F = \rho L^2 V^2 \phi\left[(\frac{M}{SVL})(\frac{Lg}{V^2})\right]$. **09**

   c. Define Reynold's number. **02**

7. a. With a neat sketch define: Laminar boundary layer, turbulent boundary layer, Laminar sub- Layer. Boundary Layer thickness. **07**

   b. What are the different methods of preventing the separation of boundary layers? **07**

8. a. A horizontal venturimeter with inlet diameter 20cm and throat diameter 10cm is used to measure the flow of oil of specific gravity 0.8. The discharge of oil through venturimeter is 60 litres/sec. Find the reading of the oil- mercury differential manometer. Take $C_d = 0.98$. **07**

   b. The head of water over an orifice of diameter 100mm is 10m. The water coming out from orifice is collected in a circular tank os diameter 1.5m. The rise of water level in this tank is 1.0m in 25 seconds. Also the Co-ordinates of a point on the jet, measured from vena-contracta are 4.3m horizontal and 0.5m vertical. Find the hydraulic co-efficients, C_d, C_v and C_c of the orifice. **07**